Health watch — sensor board

Sơ đồ phân cấp: mỗi ô là một trang con. Nhãn toàn cục (global label) nối tín hiệu giữa các trang; ký hiệu nguồn nối theo tên.

Ref 201–299. Sinh tự động bởi tools/gen_sch.py từ tools/design.py.

PPG AFE, LEDs, photodiodes, level shifter, skin temperature

File: health_watch_sensor_sensor.kicad_sch

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Sheet: /
File: health_watch_sensor.kicad_sch

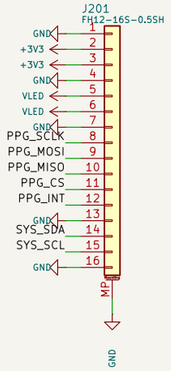
Title: Health watch — sensor board — root

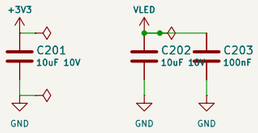
Size: A4	Date: 2026-11-05	**Rev: 1.2**
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PPG AFE, LEDs, photodiodes, level shifter, skin temperature

Bo cảm biến: LDO 1.8 V nhiều thấp → MAX86141, LED SFH 7016 (xanh/đỏ/IR) và 2 photodiode nối thẳng vào AFE; SPI 1.8 V ↔ 3.3 V qua TXS0108E; MAX30208 nhiệt độ da.

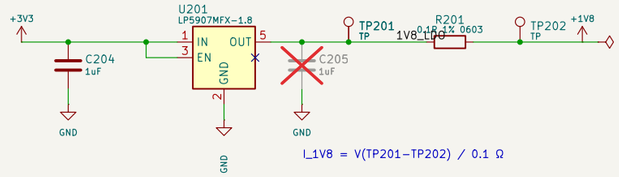
FPC 16P nối bo chính



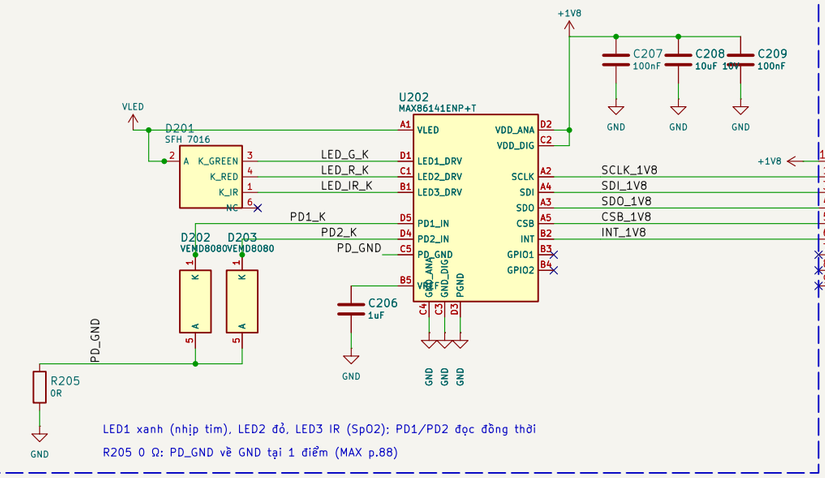


C202 10 µF sát VLED (MAX p.12)

LDO 1.8 V + shunt đo dòng



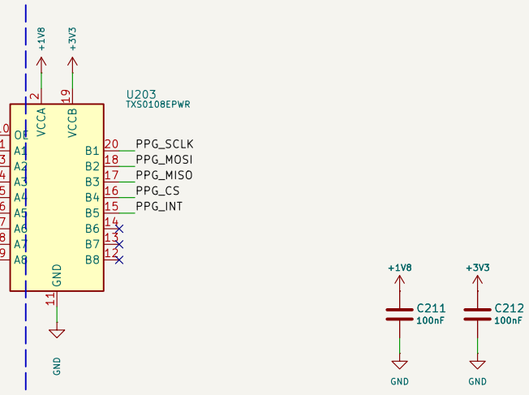
MAX86141 + LED + photodiode



LED1 xanh (nhịp tim), LED2 đỏ, LED3 IR (SpO2); PD1/PD2 đọc đồng thời

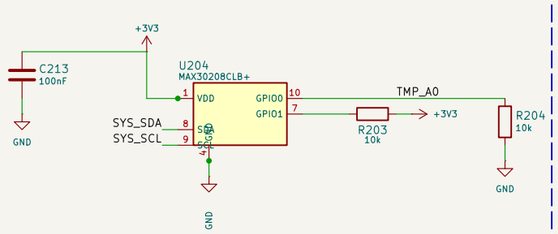
R205 0 Ω: PD_GND về GND tại 1 điểm (MAX p.88)

Chuyển mức TXS0108E (1.8 ↔ 3.3 V)



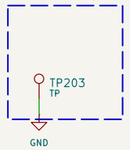
OE nối thẳng VCCA (TXS p.6: OE Δt/Δv ≤ 10 ns/V)

MAX30208 nhiệt độ da (I2C 0x52)



GPIO1 = 1, GPIO0 = 0 → địa chỉ 0x52

Điểm đo / linh kiện phụ



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Sheet: /PPG AFE, LEDs, photodiodes, level shifter, skin temperature/

File: health_watch_sensor_sensor.kicad_sch

Title: Health watch — sensor board — PPG AFE, LEDs, photodiodes, level shif

Size: A3	Date: 2026-11-05	**Rev: 1.2**
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